

AERO 434

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Outline

- Introduction of Laplace Transform (LT)
- Properties and definitions
- How to solve ODEs using LT
- Understand partial fraction expansion
- Reading sections: 2.4 Dorf and Bishop -Examples

Laplace Transform: Basic Definition

- **Laplace Transform:** takes a function of t (time) to a function of a complex variable s (frequency).
- Given a function in time ($t \geq 0$), $f(t)$, we want to apply the integral transformation:

$$F(s) = \mathcal{L}\{f(t)\} = \int_0^{\infty} f(t) e^{-st} dt$$

- The above integral might be undefined for some s .
- **Abcissa of convergence:** smallest σ such that for all s with $\text{Re}(s) > \sigma$, the integral above converges.

Example 1: Step, Ramp, and Exponential Signals

1. Laplace Transform of a Step Function

$f(t) = 5$ for all $t \geq 0$

$$\int_0^{\infty} 5e^{-st} dt = \left[-\frac{5}{s} e^{-st} \right]_0^{\infty} = \frac{5}{s} \quad \text{if } \operatorname{Re}(s) > 0$$

2. Laplace Transform of a Ramp Function

$f(t) = 2t$ for all $t \geq 0$

$$\int_0^{\infty} 2te^{-st} dt = \left[-\frac{2(1+st)}{s^2} e^{-st} \right]_0^{\infty} = \frac{2}{s^2} \quad \text{if } \operatorname{Re}(s) > 0$$

3. Laplace Transform of $f(t) = e^{-at}$ for all $t \geq 0$:

$$\int_0^{\infty} e^{-at} e^{-st} dt = \int_0^{\infty} e^{-(s+a)t} dt = \left[-\frac{e^{-(s+a)t}}{s+a} \right]_0^{\infty} = \frac{1}{s+a}$$

if:

$$\operatorname{Re}(s+a) > 0$$

Examples of Laplace Transform

- Step: $\mathcal{L}\{5\} = \frac{5}{s}$
- Ramp: $\mathcal{L}\{2t\} = \frac{2}{s^2}$
- Exponential: $\mathcal{L}\{e^{-at}\} = \frac{1}{s+a}$

Laplace Transform Table 1

	$f(t)$	$F(s)$
1	Unit impulse $\delta(t)$	1
2	Unit step $1(t)$	$\frac{1}{s}$
3	t	$\frac{1}{s^2}$
4	$\frac{t^{n-1}}{(n-1)!} \quad (n = 1, 2, 3, \dots)$	$\frac{1}{s^n}$
5	$t^n \quad (n = 1, 2, 3, \dots)$	$\frac{n!}{s^{n+1}}$
6	e^{-at}	$\frac{1}{s+a}$
7	te^{-at}	$\frac{1}{(s+a)^2}$
8	$\frac{1}{(n-1)!} t^{n-1} e^{-at} \quad (n = 1, 2, 3, \dots)$	$\frac{1}{(s+a)^n}$
9	$t^n e^{-at} \quad (n = 1, 2, 3, \dots)$	$\frac{n!}{(s+a)^{n+1}}$
10	$\sin at$	$\frac{\omega}{s^2 + \omega^2}$
11	$\cos at$	$\frac{s}{s^2 + \omega^2}$
12	$\sinh at$	$\frac{\omega}{s^2 - \omega^2}$
13	$\cosh at$	$\frac{s}{s^2 - \omega^2}$
14	$\frac{1}{a}(1 - e^{-at})$	$\frac{1}{s(s+a)}$
15	$\frac{1}{b-a}(e^{-at} - e^{-bt})$	$\frac{1}{(s+a)(s+b)}$
16	$\frac{1}{b-a}(be^{-bt} - ae^{-at})$	$\frac{s}{(s+a)(s+b)}$
17	$\frac{1}{ab} \left[1 + \frac{1}{a-b}(be^{-at} - ae^{-bt}) \right]$	$\frac{1}{s(s+a)(s+b)}$

18	$\frac{1}{a^2}(1 - e^{-at} - ate^{-at})$	$\frac{1}{s(s+a)^2}$
19	$\frac{1}{a^2}(at - 1 + e^{-at})$	$\frac{1}{s^2(s+a)}$
20	$e^{-at} \sin at$	$\frac{\omega}{(s+a)^2 + \omega^2}$
21	$e^{-at} \cos at$	$\frac{s+a}{(s+a)^2 + \omega^2}$
22	$\frac{\omega_\xi}{\sqrt{1-\xi^2}} e^{-\zeta\omega_\xi t} \sin \omega_\xi \sqrt{1-\xi^2} t \quad (0 < \xi < 1)$	$\frac{\omega_\xi^2}{s^2 + 2\xi\omega_\xi s + \omega_\xi^2}$
23	$-\frac{1}{\sqrt{1-\xi^2}} e^{-\zeta\omega_\xi t} \sin(\omega_\xi \sqrt{1-\xi^2} t - \phi)$ $\phi = \tan^{-1} \frac{\sqrt{1-\xi^2}}{\xi}$ $(0 < \xi < 1, \quad 0 < \phi < \pi/2)$	$\frac{s}{s^2 + 2\xi\omega_\xi s + \omega_\xi^2}$
24	$1 - \frac{1}{\sqrt{1-\xi^2}} e^{-\zeta\omega_\xi t} \sin(\omega_\xi \sqrt{1-\xi^2} t + \phi)$ $\phi = \tan^{-1} \frac{\sqrt{1-\xi^2}}{\xi}$ $(0 < \xi < 1, \quad 0 < \phi < \pi/2)$	$\frac{\omega_\xi^2}{s(s^2 + 2\xi\omega_\xi s + \omega_\xi^2)}$
25	$1 - \cos at$	$\frac{\omega^2}{s(s^2 + \omega^2)}$
26	$at - \sin at$	$\frac{\omega^3}{s^2(s^2 + \omega^2)}$
27	$\sin at - at \cos at$	$\frac{2\omega^3}{(s^2 + \omega^2)^2}$
28	$\frac{1}{2\omega} t \sin at$	$\frac{s}{(s^2 + \omega^2)^2}$
29	$t \cos at$	$\frac{s^2 - \omega^2}{(s^2 + \omega^2)^2}$
30	$\frac{1}{\omega_1^2 - \omega_2^2} (\cos \omega_1 t - \cos \omega_2 t) \quad (\omega_1^2 \neq \omega_2^2)$	$\frac{s}{(s^2 + \omega_1^2)(s^2 + \omega_2^2)}$
31	$\frac{1}{2\omega} (\sin at + at \cos at)$	$\frac{s^2}{(s^2 + \omega^2)^3}$

Laplace Transform Theorems

- **Linearity:** $\mathcal{L}\{\alpha_1 f_1(t) + \alpha_2 f_2(t)\} = \alpha_1 F_1(s) + \alpha_2 F_2(s)$
- **Differentiation:** $\mathcal{L}\{f'(t)\} = sF(s) - f(0)$
- **Integration:** $\mathcal{L}\left\{\int_0^t f(\tau)d\tau\right\} = \frac{F(s)}{s}$

Item no.	Theorem	Name
1.	$\mathcal{L}[f(t)] = F(s) = \int_{0-}^{\infty} f(t)e^{-st}dt$	Definition
2.	$\mathcal{L}[kf(t)] = kF(s)$	Linearity theorem
3.	$\mathcal{L}[f_1(t) + f_2(t)] = F_1(s) + F_2(s)$	Linearity theorem
4.	$\mathcal{L}[e^{-at}f(t)] = F(s+a)$	Frequency shift theorem
5.	$\mathcal{L}[f(t-T)] = e^{-sT}F(s)$	Time shift theorem
6.	$\mathcal{L}[f(at)] = \frac{1}{a}F\left(\frac{s}{a}\right)$	Scaling theorem
7.	$\mathcal{L}\left[\frac{df}{dt}\right] = sF(s) - f(0-)$	Differentiation theorem
8.	$\mathcal{L}\left[\frac{d^2f}{dt^2}\right] = s^2F(s) - sf(0-) - f'(0-)$	Differentiation theorem
9.	$\mathcal{L}\left[\frac{d^nf}{dt^n}\right] = s^nF(s) - \sum_{k=1}^n s^{n-k}f^{(k-1)}(0-)$	Differentiation theorem
10.	$\mathcal{L}\left[\int_{0-}^t f(\tau)d\tau\right] = \frac{F(s)}{s}$	Integration theorem
11.	$f(\infty) = \lim_{s \rightarrow 0} sF(s)$	Final value theorem ¹
12.	$f(0+) = \lim_{s \rightarrow \infty} sF(s)$	Initial value theorem ²

¹For this theorem to yield correct finite results, all roots of the denominator of $F(s)$ must have negative real parts, and no more than one can be at the origin.

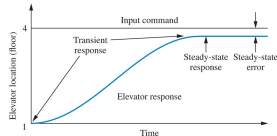
²For this theorem to be valid, $f(t)$ must be continuous or have a step discontinuity at $t = 0$ (that is, no impulses or their derivatives at $t = 0$).

Higher Order Differentiation

$$\mathcal{L}\{f^{(n)}(t)\} = s^n F(s) - s^{n-1}f(0) - \dots - sf^{(n-2)}(0) - f^{(n-1)}(0)$$

Poles, Zeros, and Final Value Theorem

- Consider: $F(s) = \frac{N(s)}{D(s)}$
- Poles:** roots of $D(s)$
- Zeros:** roots of $N(s)$



$$\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} sF(s), \quad \text{if all poles of } sF(s) \text{ are in LHP.}$$

Example 1:

$$F(s) = \frac{5}{s(s^2 + s + 2)}, \quad \lim_{t \rightarrow \infty} f(t) = \left(\frac{5}{2}\right) \text{ (steady-state response)}$$

Example 2:

$$F(s) = \frac{4}{s^2 + 4}, \quad \lim_{t \rightarrow \infty} f(t) = 0 \quad \text{(WRONG!)}$$

The theorem fails here since poles at $s = \pm 2j$ lie on the imaginary axis.

FVT Example Problem

ODE: $\ddot{y}(t) + 3\dot{y}(t) + 2y(t) = 2\dot{u}(t) + u(t)$

All initial conditions zero, $u(t) = 1$.

Laplace transform:

$$s^2 Y(s) + 3sY(s) + 2Y(s) = 2sU(s) + U(s)$$

For $u(t) = 1$, $U(s) = \frac{1}{s}$:

$$s^2 Y(s) + 3sY(s) + 2Y(s) = 2 + \frac{1}{s}$$

$$Y(s)[s^2 + 3s + 2] = 2 + \frac{1}{s}$$

$$Y(s) = \frac{2 + \frac{1}{s}}{s^2 + 3s + 2} = \frac{2s + 1}{s(s^2 + 3s + 2)}$$

$$\lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow 0} sY(s) = \lim_{s \rightarrow 0} \frac{2s + 1}{s^2 + 3s + 2} = \frac{1}{2} \text{ (steady-state response)}$$

Laplace Transform Properties: Initial Value Theorem

- Consider: $F(s) = \frac{N(s)}{D(s)}$

$$\lim_{t \rightarrow 0^+} f(t) = \lim_{s \rightarrow \infty} sF(s), \quad \text{if the limit exists}$$

- Note:** In this theorem, it does not matter if the pole location is in LHP (Left Half Plane) or not.

Example 1:

$$F(s) = \frac{5}{s(s^2 + s + 2)}, \quad \lim_{t \rightarrow 0^+} f(t) = \lim_{s \rightarrow \infty} sF(s) = \lim_{s \rightarrow \infty} \frac{5}{s^2 + s + 2} = 0$$

Example 2:

$$F(s) = \frac{4}{s^2 + 4}, \quad \lim_{t \rightarrow 0^+} f(t) = \lim_{s \rightarrow \infty} sF(s) = \lim_{s \rightarrow \infty} \frac{4s}{s^2 + 4} = 0$$

Laplace Transform Properties: Convolution

- **What is convolution?**

$$f_1(t) * f_2(t) = \int_0^t f_1(t - \tau)f_2(\tau)d\tau = \int_0^t f_2(t - \tau)f_1(\tau)d\tau$$

- A very powerful tool in probability, dynamics, image processing, and signal processing, but often requires significant computation.

$$\mathcal{L}\{f_1(t) * f_2(t)\} = \mathcal{L}\{f_1(t)\} \cdot \mathcal{L}\{f_2(t)\} = F_1(s)F_2(s)$$

- **Note:** $F_1(s)F_2(s) \neq \mathcal{L}\{f_1(t)f_2(t)\}$ (Product in Laplace is convolution in time.)
- Laplace makes convolution computations much simpler and avoids lengthy integration.

Laplace Transform Properties: Inverse Laplace Transform

- What if you have $F(s)$? Can you obtain $f(t)$ back? **Yes.**

$$f(t) = \mathcal{L}^{-1}\{F(s)\} = \frac{1}{2\pi j} \int_{c-j\infty}^{c+j\infty} F(s)e^{st} ds$$

- The above integral is often very hard to solve directly.
- In practice, we usually look up tables of Laplace transform pairs to find the answer, or use **partial fraction expansion** to simplify $F(s)$ before using the table.

Alternative: Partial fraction expansion makes finding the inverse laplace much simpler!

Partial Fraction Expansion (PFE)

- **Main Objective:** Find the inverse Laplace transform of $F(s)$ given:

$$F(s) = \frac{N(s)}{D(s)} = \frac{b_0 s^m + b_1 s^{m-1} + \dots + b_m}{a_0 s^n + a_1 s^{n-1} + \dots + a_n} \quad n \geq m$$

- For linear systems, $N(s)$ and $D(s)$ are always polynomials.
- Typically, order $D(s) \geq$ order $N(s)$ (i.e., $n \geq m$).
- **Secondary Objective:** Write $F(s) = \frac{N(s)}{D(s)}$ in terms of known expressions from the LT table.

Three major cases for the roots of $D(s)$ in PFE:

- A) Distinct roots
- B) Equal/repeated roots
- C) Complex conjugate roots

Case A: Distinct Roots of $D(s)$

- If all roots $p_1, p_2, \dots, p_n$ of the denominator are distinct:

$$F(s) = \frac{N(s)}{D(s)} = \frac{b_0 s^m + b_1 s^{m-1} + \dots + b_m}{a_0 (s - p_1)(s - p_2) \dots (s - p_n)}$$

- Since $n \geq m$, we can write:

$$F(s) = \frac{c_1}{s - p_1} + \frac{c_2}{s - p_2} + \dots + \frac{c_n}{s - p_n}$$

- How will this form be useful to find $f(t)$?
- That's easy: $f(t) = \mathcal{L}^{-1} [F(s)] = c_1 e^{p_1 t} + \dots + c_n e^{p_n t}$
- How to find residue? Assume that p_i 's are given, then find the so-called residue c_i 's for corresponding pole p_i 's

$$c_i = (s - p_i)F(s) \Big|_{s=p_i}$$

EXAMPLE of Distinct Roots: Inverse Laplace Transform

Find the inverse Laplace transform of:

$$F(s) = \frac{s+3}{(s+1)(s+2)}$$

The partial-fraction expansion for $F(s)$ is:

$$\frac{s+3}{(s+1)(s+2)} = \frac{c_1}{s+1} + \frac{c_2}{s+2}$$

where the coefficients c_1 and c_2 are found as:

$$c_1 = [(s+1)F(s)]_{s=-1} = [(s+3)/(s+2)]_{s=-1} = 2$$

$$c_2 = [(s+2)F(s)]_{s=-2} = [(s+3)/(s+1)]_{s=-2} = -1$$

Thus,

$$f(t) = \mathcal{L}^{-1}[F(s)] = \mathcal{L}^{-1}\left[\frac{2}{s+1}\right] + \mathcal{L}^{-1}\left[\frac{-1}{s+2}\right]$$

$$f(t) = 2e^{-t} - e^{-2t}, \quad t \geq 0$$

```
num = [1 3]; den = [1 3 2]; [r, p, k] = residue(num, den)
```

EXAMPLE of Repeated Roots: Inverse Laplace Transform

Find the inverse Laplace transform of:

$$F(s) = \frac{s^2 + 2s + 3}{(s + 1)^3}$$

The partial-fraction expansion of this $F(s)$ involves three terms:

$$F(s) = \frac{B(s)}{A(s)} = \frac{b_1}{s + 1} + \frac{b_2}{(s + 1)^2} + \frac{b_3}{(s + 1)^3}$$

where b_3 , b_2 , and b_1 are determined as follows. By multiplying both sides of this last equation by $(s + 1)^3$, we have

$$(s + 1)^3 \frac{B(s)}{A(s)} = b_1(s + 1)^2 + b_2(s + 1) + b_3 \quad (1)$$

Thus letting $s = -1$, Equation (1) becomes

$$\left[(s + 1)^3 \left(\frac{B(s)}{A(s)} \right) \right]_{s=-1} = b_3$$

EXAMPLE of Repeated Roots- (continuation ...)

Also, differentiation of both sides of the above equation with respect to s yields

$$\frac{d}{ds} \left[(s+1)^3 \frac{B(s)}{A(s)} \right] = b_2 + 2b_1(s+1) \quad (2)$$

If we let $s = -1$ in equation (2), then

$$\frac{d}{ds} \left[(s+1)^3 \frac{B(s)}{A(s)} \right]_{s=-1} = b_2$$

By differentiating both sides of Equation (2) with respect to s , the result is

$$\frac{d^2}{ds^2} \left[(s+1)^3 \frac{B(s)}{A(s)} \right] = 2b_1$$

From the preceding analysis it can be seen that the values of b_3 , b_2 , and b_1 are found systematically as follows:

$$b_3 = \left[(s+1)^3 \frac{B(s)}{A(s)} \right]_{s=-1} = (s^2 + 2s + 3)_{s=-1} = 2$$

EXAMPLE of Repeated Roots- (continuation)

$$b_2 = \left. \frac{d}{ds} \left[(s+1)^3 \frac{B(s)}{A(s)} \right] \right|_{s=-1} = \left. \frac{d}{ds} (s^2 + 2s + 3) \right|_{s=-1} = (2s + 2)|_{s=-1} = 0$$

$$b_1 = \frac{1}{2!} \left. \frac{d^2}{ds^2} \left[(s+1)^3 \frac{B(s)}{A(s)} \right] \right|_{s=-1} = \frac{1}{2!} \left. \frac{d^2}{ds^2} (s^2 + 2s + 3) \right|_{s=-1} = \frac{1}{2!} (2) = 1$$

$$f(t) = \mathcal{L}^{-1}[F(s)] = \mathcal{L}^{-1} \left[\frac{1}{s+1} \right] + \mathcal{L}^{-1} \left[\frac{0}{(s+1)^2} \right] + \mathcal{L}^{-1} \left[\frac{2}{(s+1)^3} \right]$$

$$f(t) = e^{-t} + 0 + t^2 e^{-t} = (1 + t^2) e^{-t}, \quad t \geq 0$$

```
num = [1 2 3]; den = [1 3 3 1]; [r, p, k] = residue(num,  
den)
```

EXAMPLE: Inverse Laplace Transform (Complex Poles)

Given:

$$F(s) = \frac{2s + 12}{s^2 + 2s + 5}$$

The denominator polynomial can be factored as:

$$s^2 + 2s + 5 = (s + 1 + j2)(s + 1 - j2)$$

If the function $F(s)$ involves a pair of complex-conjugate poles, it is convenient **not to expand $F(s)$ into the partial fractions** but to expand it into the **sum of a damped sine and a damped cosine** function. Noting that

$$s^2 + 2s + 5 = (s + 1)^2 + 2^2$$

and referring to the Laplace transforms of $e^{-\alpha t} \sin(\omega t)$ and $e^{-\alpha t} \cos(\omega t)$, rewritten thus:

$$\mathcal{L} \{ e^{-\alpha t} \sin(\omega t) \} = \frac{\omega}{(s + \alpha)^2 + \omega^2}$$

EXAMPLE: Complex Poles (continuation ...)

$$\mathcal{L}\{e^{-\alpha t} \cos(\omega t)\} = \frac{(s + \alpha)}{(s + \alpha)^2 + \omega^2}$$

and given $F(s)$ can be written as a sum of a damped sine and a damped cosine function

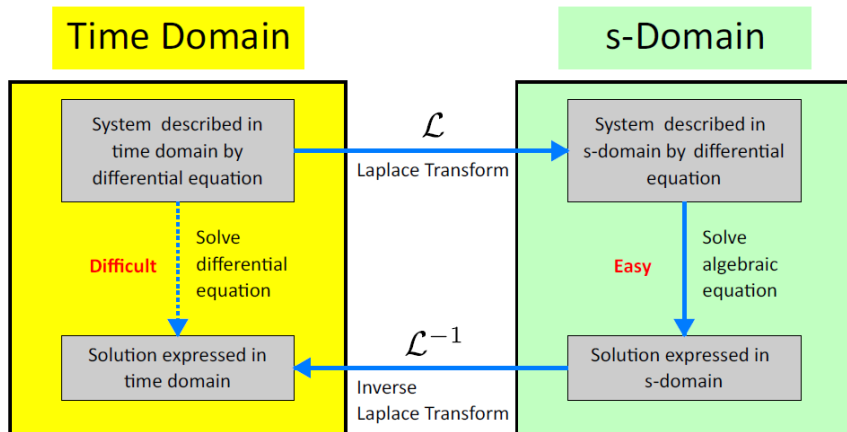
$$F(s) = \frac{2s + 12}{s^2 + 2s + 5} = \frac{10 + 2(s + 1)}{(s + 1)^2 + 2^2} = 5 \left(\frac{2}{(s + 1)^2 + 2^2} \right) + 2 \left(\frac{s + 1}{(s + 1)^2 + 2^2} \right)$$

$$f(t) = \mathcal{L}^{-1}[F(s)] = 5 \mathcal{L}^{-1} \left[\frac{2}{(s + 1)^2 + 2^2} \right] + 2 \mathcal{L}^{-1} \left[\frac{s + 1}{(s + 1)^2 + 2^2} \right]$$

$$f(t) = 5 e^{-t} \sin 2t + 2 e^{-t} \cos 2t, \quad t \geq 0$$

Why? Why Not?

- Solving ODEs using time-domain techniques can be very challenging.
- Laplace transforms and partial fraction expansions (PFEs) offer an easy and systematic way to solve ODEs.



Example 1: Solving an ODE with Laplace Transform

Solve the following ODE (find $y(t)$):

$$\ddot{y}(t) - y(t) = t, \quad y(0) = 1, \quad \dot{y}(0) = 1$$

Laplace property:

$$\mathcal{L}\{f^{(n)}(t)\} = s^n F(s) - s^{n-1}f(0) - s^{n-2}f^{(1)}(0) - \dots - f^{(n-1)}(0)$$

Solution Steps:

- 1 Take Laplace Transform
- 2 Find $Y(s)$
- 3 Apply PFE for $Y(s)$, finding poles and residues
- 4 Use the ILT to find $y(t)$

Answer:

$$y(t) = 1.5 e^t - 0.5 e^{-t} - t$$

MATLAB: Solving ODE Step-by-Step

Step 1: Take Laplace Transform

```
syms y(t) Y s t
ode = diff(y, t, 2) - y == t;
L_ODE = laplace(ode, t, s);
```

Step 2: Find $Y(s)$ with Initial Conditions

```
L_ODE = subs(L_ODE, laplace(y, t, s), Y);
L_ODE = subs(L_ODE, [y(0), subs(diff(y, t), t, 0)], [1, 1]);
Ysol = solve(L_ODE, Y);
```

Step 3: Apply Partial Fraction Expansion

```
Ysol_pfe = partfrac(Ysol, s);
```

Step 4: Use Inverse Laplace Transform

```
y_sol = ilaplace(Ysol, s, t);
pretty(y_sol)
```

Example 2: Solving ODE with Laplace Transform

Solve the following ODE (find $y(t)$):

$$\ddot{y}(t) + 3\dot{y}(t) + 2y(t) = 5, \quad y(0) = -1, \quad \dot{y}(0) = 2$$

Solution Steps:

① Take Laplace Transform:

$$s^2 Y(s) - sy(0) - \dot{y}(0) + 3[sY(s) - y(0)] + 2Y(s) = \frac{5}{s}$$

② Substitute initial conditions ($y(0) = -1, \dot{y}(0) = 2$):

$$s^2 Y(s) + s - 2 + 3[sY(s) + 1] + 2Y(s) = \frac{5}{s}$$

③ Collect $Y(s)$ terms:

$$(s^2 + 3s + 2)Y(s) + (s + 1) = \frac{5}{s}$$

Example 2: Solving ODE with Laplace Transform (continuation ..)

Rearrange and solve:

$$(s^2 + 3s + 2)Y(s) = \frac{5}{s} - (s + 1)$$

$$Y(s) = \frac{5}{s(s^2 + 3s + 2)} - \frac{s + 1}{s^2 + 3s + 2}$$

- 4 Apply Partial Fraction Expansion to $Y(s)$, finding poles and residues.
- 5 Use Inverse Laplace Transform (ILT) to find $y(t)$.
- 6 **Answer:**

$$y(t) = 1.5 e^{-2t} - 5e^{-t} + 2.5$$

MATLAB Demo: Laplace and Symbolic Toolbox

- Most of what we learned today can be easily tested on MATLAB.
- **Symbolic Toolbox:** provides functions for solving and manipulating symbolic math expressions.

Symbolic variable definition:

```
syms s t
```

To find the Laplace transform, use:

```
laplace
```

To find the inverse Laplace transform, use:

```
ilaplace
```

Modeling

- Modeling of Systems (ODEs)
- Linearization
- Solution of ODEs
- Laplace Transforms
- Block Diagrams
- SISO/MIMO analysis

Analysis

- The 1st and 2nd order systems
- Time response
- Transient/Steady state response
- RH Criterion
- Stability
- Frequency Response
- Bode Plots

Design

- Root-Locus
- Modern Control
- State-Space
- MIMO system properties